1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist's path indicates PROS1 is expressed in tumors and is associated with Ki-67, suggesting PROS1 correlates with proliferation; the puzzle is why tumors with high stromal content (typically thought to restrain tumor cells) might nonetheless display high proliferative indices. 2) Tier 1 paths used: (a) 'tumors → is expressed in → PROS1' and (b) 'PROS1 → is associated with → Ki-67 (MKI67)'. I built on these because they provide the direct molecular link between a tumor-expressed coagulation regulator and proliferation. 3) Tier 2 supporting evidence: 'gene expression → analyzing → tumors' supports that transcriptional measurements can resolve PROS1 expression, and the relationship noting ultrasound does not alter gene expression reduces concern about imaging-induced transcriptional artifacts; the PROS1 definition (tumor–host/coagulation interactions) provides plausible mechanistic routes by which PROS1 could modulate the microenvironment to affect proliferation. 4) Endpoint choice rationale: I selected progression-free interval (PFI) because MKI67-linked proliferative activity is mechanistically and clinically most proximate to disease progression risk rather than long-term mortality, making PFI the most sensitive endpoint to test a PROS1–proliferation effect. 5) Deliberate exclusions: I excluded the ultrasound/bladder nodes from the core hypothesis (they are methodological/contextual) and omitted other coagulation or stroma-related genes not present in the Ontologist's Tier 1 to keep the hypothesis tightly focused on the PROS1–MKI67 axis; no additional genes were forced into the main claim to satisfy the Focusing Rule?

Working Hypothesis

In breast tumors, high stromal-to-tumor ratio (TSR) combined with elevated intratumoral PROS1 expression drives a pro-proliferative stromal microenvironment that increases MKI67 expression in tumor cells through PROS1-related signaling within the stroma. Clinically, this PROS1–proliferation axis predicts shorter progression-free interval (PFI) in patients with TSR-high tumors, whereas TSR-low tumors show a weaker or absent PROS1–MKI67 association and no reduction in PFI.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PROS1, MKI67
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PROS1: PROS1 expression, and the relationship noting ultrasound does not alter gene expression reduces concern about imaging-induced transcriptional artifacts; the PROS1 definition (tumor–host/coagulation interactions) provides plausible mechanistic routes by which PROS1 could modulate the microenvironment to affect proliferation.

MKI67: MKI67-linked proliferative activity is mechanistically and clinically most proximate to disease progression risk rather than long-term mortality, making PFI the most sensitive endpoint to test a PROS1–proliferation effect.

3. Clinically Testable Predictions

Prediction 1: stratified correlation of PROS1 and MKI67 by TSR;

Prediction 2: multivariable models for PFI incorporating PROS1, MKI67, TSR, and their interaction;

Prediction 3: mediation/moderation analyses to assess whether PROS1 mediates the TSR–MKI67 relationship and whether this predicts PFI;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions